

products.csv – Data Exploration Report (In Progress)

Exploration of `products.csv` : structural profile → null/missingness pass → duplicate pass → column-specific notes. Covers `asin` , `availability` , `best_sellers_rank` , `brand_name` , `brand_page_url` , `breadcrumbs` , and `default_variant/0-2` . Remaining columns and referential integrity against `reviews.csv` are pending a follow-up session. Full working evidence is in `notebooks/products_exploration.ipynb` .

Structural Profile

- Shape: 728 rows × 34 columns.

Null / Missingness Pass

Column	Null %
s.no	0.0
about_item	0.0
asin	0.0
availability	1.79
best_sellers_rank	23.35
brand_name	0.0
brand_page_url	10.85
breadcrumbs	1.79
customer_review_summary	12.91
default_variant/0	4.4
default_variant/1	9.62
default_variant/2	99.86

Column	Null %
delivery_date	3.57
fastest_delivery_date	8.38
list_price	44.92
manufacturer	63.46
model_number	72.66
price_value	2.88
product_description	62.77
product_url	0.0
rating_count	1.24
rating_distribution/1star-5star	0.0
rating_stars	1.24
recent_purchases	21.84
scrape_time	0.0
seller_name	2.88
seller_page_url	41.48
title	0.0
all_images	0.0
rank_1	25.82

Duplicate Pass

- `asin` duplicates: 0 — fully unique across all 728 rows. Safe key for a `products` collection.

Column-Specific Notes

`availability`

Several issues need unifying:

1. `In Stock` and `In stock` are the same value, differing only by capitalization – needs normalizing in the Transformer.
2. Almost all values will need to collapse to one of 2 options – in stock or unavailable – or roughly 3: in stock, unavailable, and limited.

`best_sellers_rank`

Contains the product's rank in the best sellers list. What matters is the numeric rank itself.

For the Transformer:

1. Extract the numbers and provide a new column for the category.
2. Handle the number string → int conversion cleanly.

`brand_name`

1. 79 brand names contain "Store" in them, needing removal with the pattern `"xxx Store"`.
2. The rest contain the word "Brand" at the start, with the pattern `"Brand: xxx"`.
3. One value is literally the string `"Unknown"`, which should be treated as NaN/null – to handle in the Transformer.

`brand_page_url`

No suspicious/malformed links found – anomaly check only surfaced nulls (79 rows), no broken URL structures. Safe to use as-is.

`breadcrumbs`

1. No need to do heavy parsing work on `best_sellers_rank` for category extraction, since `breadcrumbs` already provides the category via a clean separator (the `>` trail).
2. When `breadcrumbs` is null, `best_sellers_rank` is also null – but not the other way around (157 rows have null `best_sellers_rank` with a populated `breadcrumbs`). This rules out using `breadcrumbs` to compensate for missing `best_sellers_rank` values.

`default_variant/0, /1, /2`

Most of `default_variant/0` holds the size of the variant, with a small amount of pollution (11 rows) containing "color" instead. `default_variant/1` is the mirror case – mostly color, with 1 row of pollution. `default_variant/2` is 99.86% null.

Next Steps

Remaining columns not yet profiled: `customer_review_summary`, `delivery_date`, `fastest_delivery_date`, `list_price`, `manufacturer`, `model_number`, `price_value`, `product_description`, `product_url`, `rating_count`, `rating_distribution/1star-5star`, `rating_stars`, `recent_purchases`, `scrape_time`, `seller_name`, `seller_page_url`, `title`, `all_images`, `rank_1`.

Referential integrity against `reviews.csv` (orphan `asin` values, reviews-per-product cardinality) deferred to a joint pass once both files are fully profiled.